

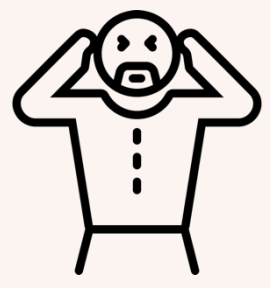


Helplessness in Dynamic Causal Environments: The Impact of Controllability and Reward Prevalence

Joshua Hew (J.W.Z.Hew@sms.ed.ac.uk), Neil Bramley

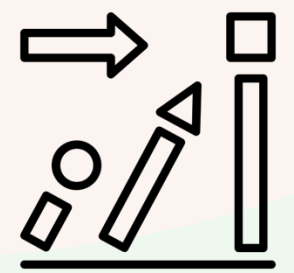
Department of Psychology, University of Edinburgh

Introduction



Learned Helplessness (LH)

Failure to avoid punishment or obtain rewards even though they are under the agent's control



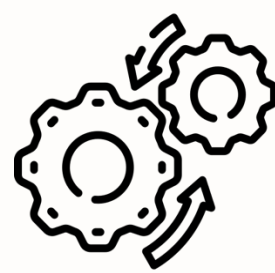
Causality in Time

Experience world in time:
E.g. Delayed effects of actions,
actions prevent/generate effects



Mechanism

Does LH arise due to insufficient prevalence of rewards or perceived incontingency between actions and outcomes?

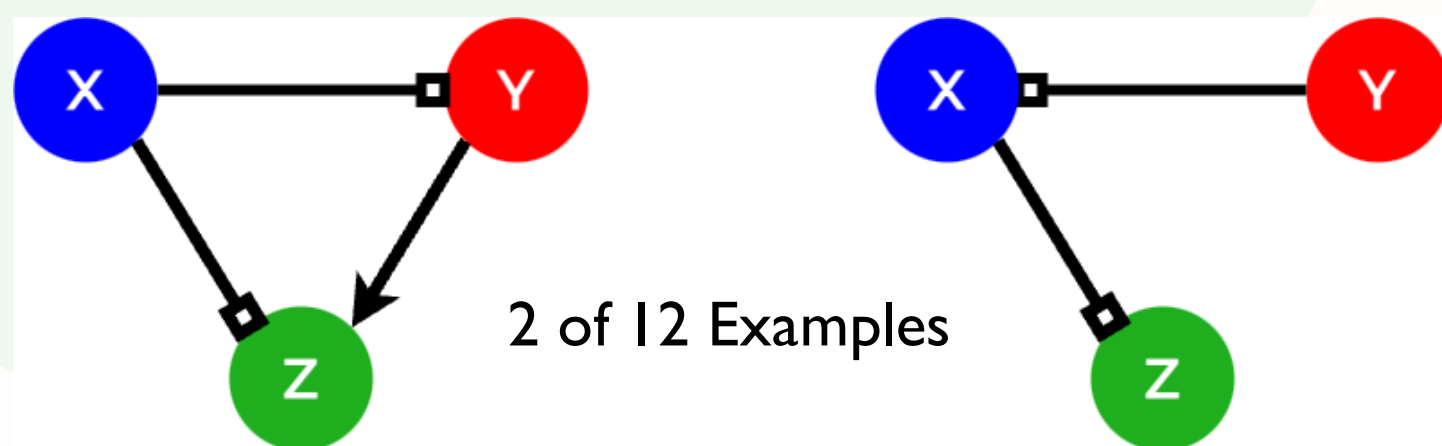


Dynamic Control

Interacting with the world requires us to leverage our knowledge of dynamic cause-effect relations

Methods

- Between-subjects manipulation (n = 48 currently):
 - High or low reward region
 - More or less noise added
- Three training rounds and three test rounds
- Three variables are related by unknown causal relationships:



2 of 12 Examples

- Goal:** Score as many points as possible by controlling sliders X and Y so that **variable Z** is in the **reward region**
- Interventions:** Movement of slider in each time step (30 total)
- Cost-reward:** At each time step, performing an intervention costs 1 point, score 3 points if **variable Z** is in the region

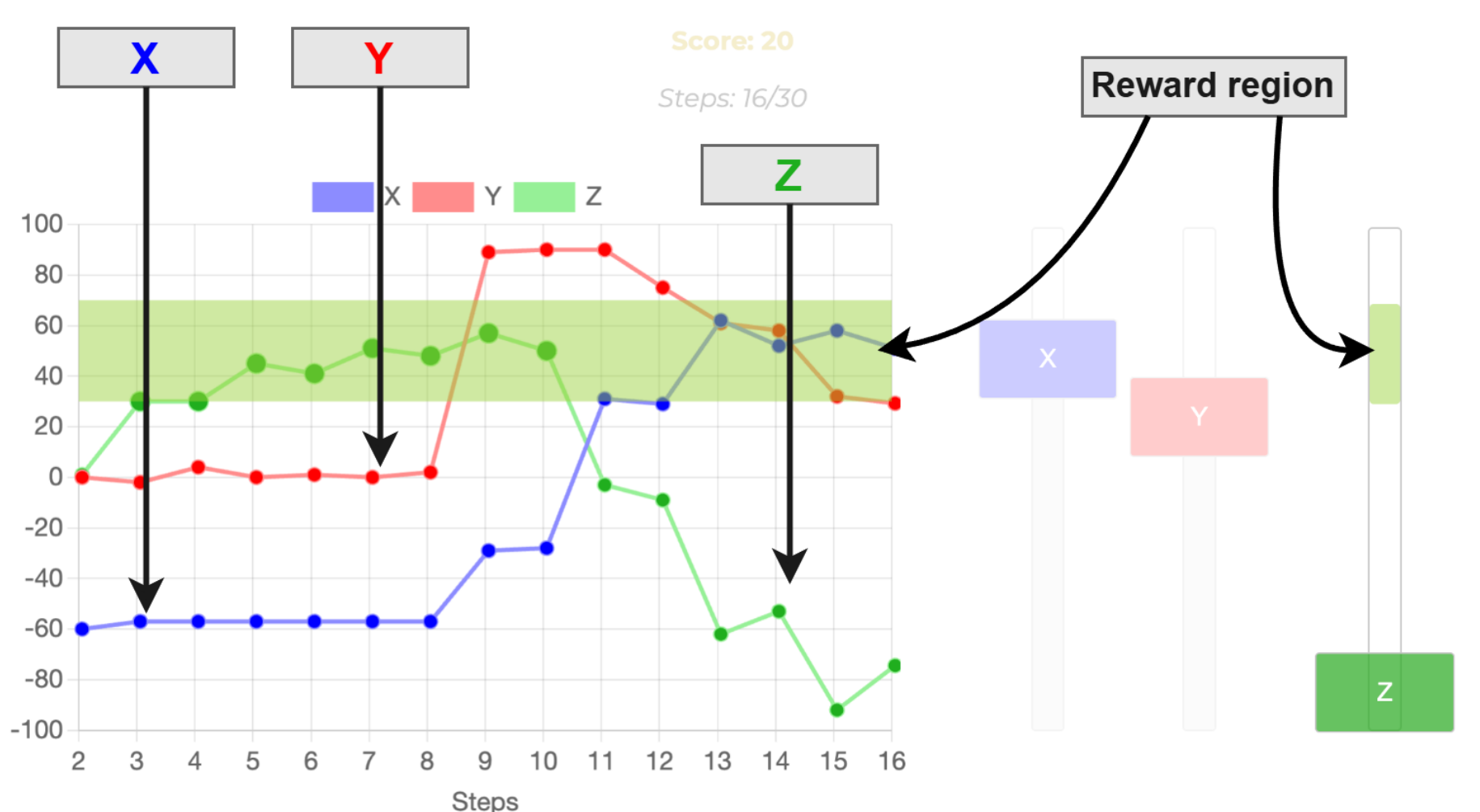


Figure 1: Annotated Task Interface

Discussion

- How to better quantify helpless behaviour in a dynamic control paradigm?
- How to quantify participant's rate of "giving up" when they drop out between training and test rounds (anecdotal evidence)

Exploratory Results

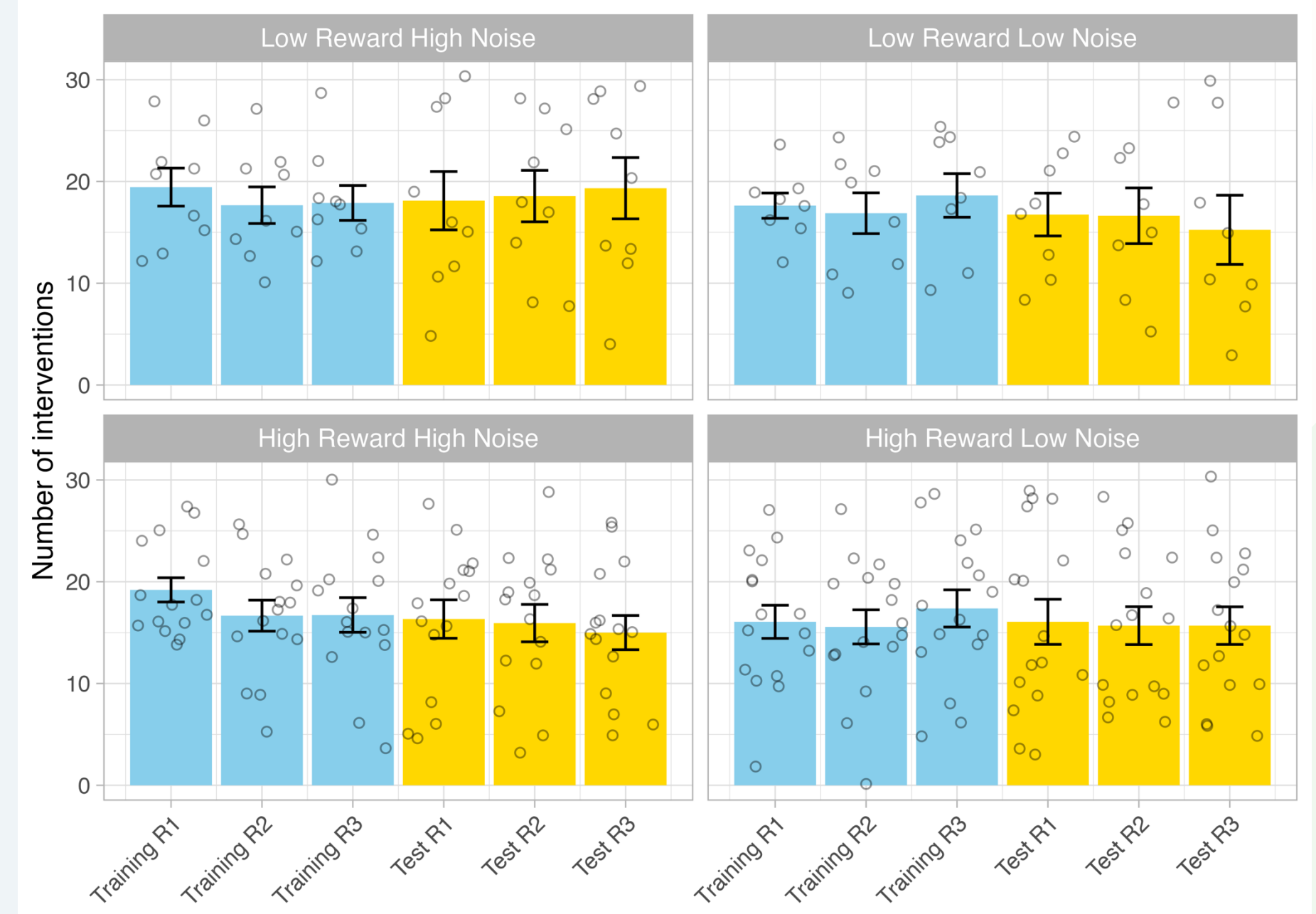


Figure 2: Number of interventions across training and test rounds

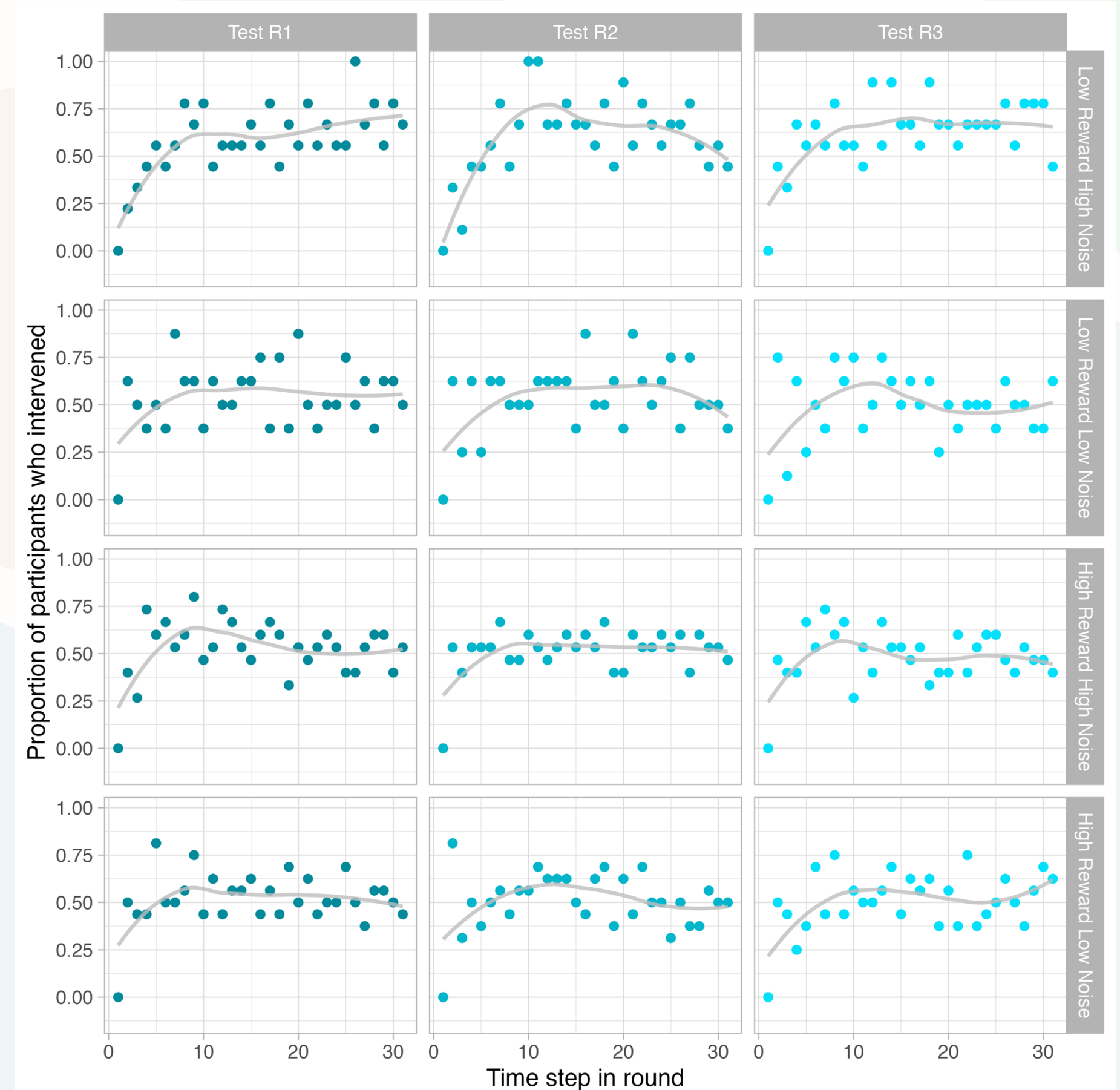


Figure 3: Pattern of interventions across test rounds

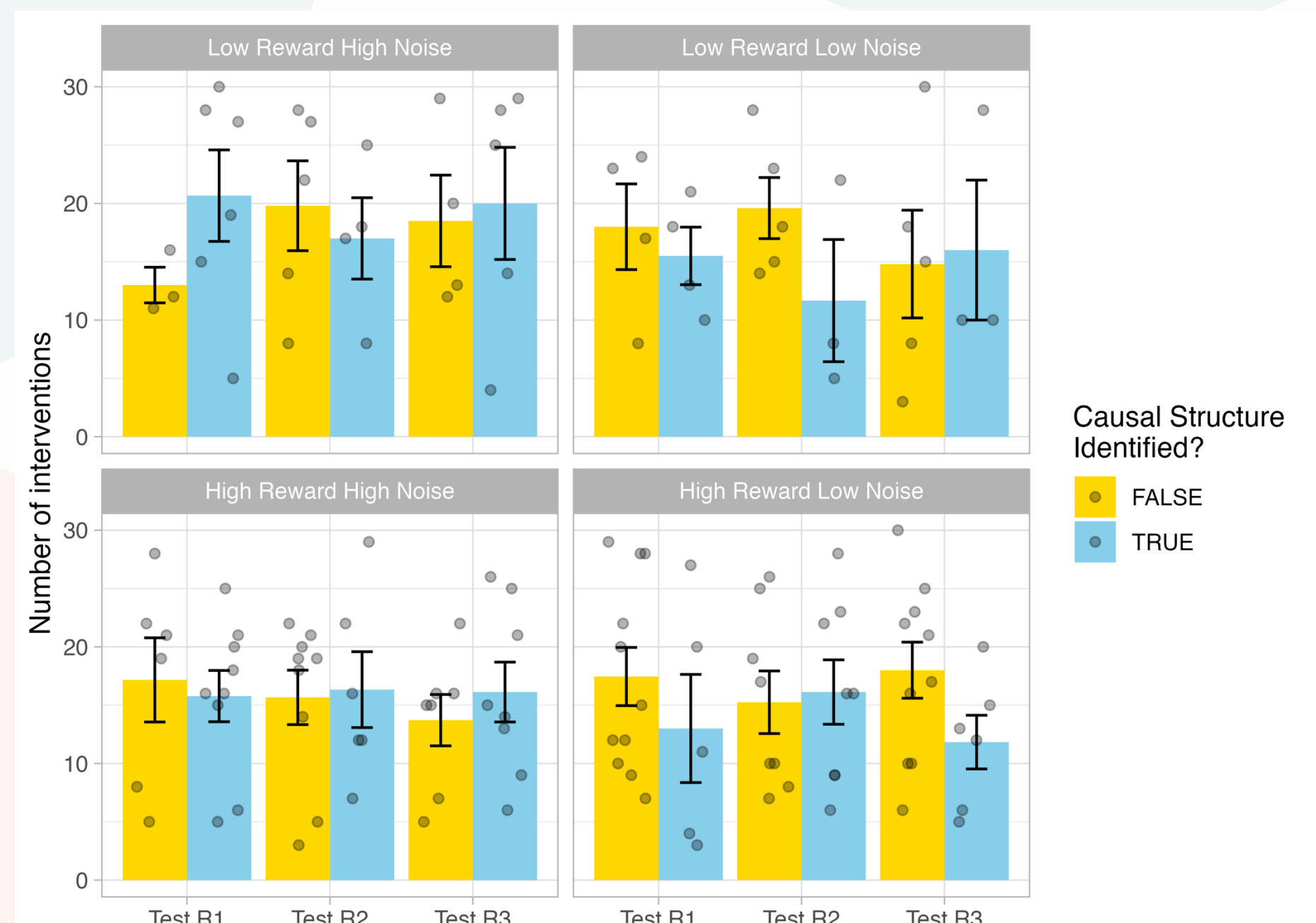


Figure 4: Number of interventions when a causal structure is identified

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